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Organic light emitting diode and organic light emitting device including the same

Abstract

The present disclosure relates to an OLED that includes a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, and wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2005/0140275	12/2004	Park	313/506	H10K 50/125
2011/0037057	12/2010	LeCloux	549/469	C07D 307/91
2012/0126208	12/2011	Kawamura	257/E51.026	H10K 85/6576
2014/0183500	12/2013	Ikeda et al.	N/A	N/A
2015/0280161	12/2014	Song	N/A	H10K 50/131
2016/0164042	12/2015	Cho	257/40	H10K 50/131
2017/0244047	12/2016	Lee	N/A	C07C 211/54
2018/0076418	12/2017	Cho et al.	N/A	N/A
2018/0301629	12/2017	Hatakeyama et al.	N/A	N/A
2019/0207112	12/2018	Hatakeyama et al.	N/A	N/A
2019/0312207	12/2018	Hatakeyama et al.	N/A	N/A
2020/0136053	12/2019	Heo et al.	N/A	N/A
2021/0367164	12/2020	Song	N/A	H10K 85/636

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107851724	12/2017	CN	N/A
108603106	12/2017	CN	N/A
10-2011-0034977	12/2010	KR	N/A
10-2016-0069468	12/2015	KR	N/A
10-2017-0130434	12/2016	KR	N/A
2017/188111	12/2016	WO	N/A
2018/150832	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report dated Apr. 6, 2020, issued in International Patent Application No. PCT/KR2019/018262. cited by applicant

Extended European Search Report dated Jul. 28, 2022, issued in corresponding European Patent Application No. 19901689.0. cited by applicant

Office Action issued in corresponding Korea Intellectual Property Office Application No. 10-2018-0172061, dated Dec. 11, 2023. (Note: 10-2011-0034977 cited therein is already of record.). cited by applicant

Office Action issued May 23, 2024 for Chinese Patent Application No. 201980050119.0 (See English Translation) (Note: KR 10-2016-0069468 A was cited in a prior IDS.). cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to an organic light emitting diode (OLED), and more specifically, to an OLED having enhanced emitting efficiency and lifespan and an organic light emitting device including the same.

BACKGROUND ART

- (2) As requests for a flat panel display device having a small occupied area have been increased, an organic light emitting display device including an OLED has been the subject of recent research and development.
- (3) The OLED emits light by injecting electrons from a cathode as an electron injection electrode and holes from an anode as a hole injection electrode into an emitting material layer (EML), combining the electrons with the holes, generating an exciton, and transforming the exciton from an excited state to a ground state. A flexible substrate, for example, a plastic substrate, can be used as a base substrate where elements are formed. In addition, the organic light emitting display device can be operated at a voltage (e.g., 10V or below) lower than a voltage required to operate other display devices. Moreover, the organic light emitting display device has advantages in the power consumption and the color sense.
- (4) The OLED includes a first electrode as an anode over a substrate, a second electrode, which is spaced apart from and faces the first electrode, and an organic emitting layer therebetween.
- (5) For example, the organic light emitting display device may include a red pixel region, a green pixel region and a blue pixel region, and the OLED may be formed in each of the red, green and

blue pixel regions.

(6) However, the OLED in the blue pixel does not provide sufficient emitting efficiency and lifespan such that the organic light emitting display device has a limitation in the emitting efficiency and the lifespan.

DISCLOSURE OF INVENTION

Technical Problem

(7) Accordingly, the present disclosure is directed to an OLED and an organic light emitting device including the OLED that substantially obviate one or more of the problems due to the limitations and disadvantages of the related art.

(8) An object of the present disclosure is to provide an OLED having enhanced emitting efficiency and lifespan and an organic light emitting device including the same.

(9) Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the disclosure. The objectives and other advantages of the disclosure will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

Solution to Problem

(10) According to an aspect, the present disclosure provides an OLED that includes a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, and wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative.

(11) As an example, in the first emitting material layer, a weight % ratio of the first host to the second host is 1:9 to 9:1.

(12) As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 1:9 to 7:3.

(13) As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 3:7.

(14) As an example, in the first emitting material layer, the weight % ratio of the first host to the second host is 7:3.

(15) The OLED may include a single emitting part or a tandem structure of a multiple emitting parts.

(16) The tandem-structured OLED may emit blue color or white color.

(17) According to another aspect, the present disclosure provides an organic light emitting device comprising the OLED, as described above.

(18) For example, the organic light emitting device may be an organic light emitting display device or a lightening device.

(19) It is to be understood that both the foregoing general description and the following detailed description are examples and are explanatory and are intended to provide further explanation of the disclosure as claimed.

Advantageous Effects of Invention

(20) An emitting material layer of an OLED of the present disclosure includes a first host of an anthracene derivative and a second host of a deuterated anthracene derivative such that an emitting efficiency and a lifespan of the OLED and an organic light emitting device including the OLED are improved.

(21) In addition, an electron blocking layer of an OLED of the present disclosure includes an amine derivative as an electron blocking material, and a hole blocking layer of the OLED includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED and an organic light emitting device is further improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure, are incorporated in and constitute a part of this specification, illustrate implementations of the disclosure and together with the description serve to explain the principles of embodiments of the disclosure.

(2) FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device of the present disclosure.

(3) FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device according to a first embodiment of the present disclosure.

(4) FIG. 3 is a schematic cross-sectional view illustrating an OLED having a single emitting part for the organic light emitting display device according to the first embodiment of the present disclosure.

(5) FIG. 4 is a schematic cross-sectional view illustrating an OLED having a tandem structure of two emitting parts according to the first embodiment of the present disclosure.

(6) FIG. 5 is a schematic cross-sectional view illustrating an organic light emitting display device according to a second embodiment of the present disclosure.

(7) FIG. 6 is a schematic cross-sectional view illustrating an OLED for the organic light emitting display device according to the second embodiment of the present disclosure.

(8) FIG. 7 is a schematic cross-sectional view illustrating an organic light emitting display device according to a third embodiment of the present disclosure.

MODE FOR THE INVENTION

(9) Reference will now be made in detail to aspects of the disclosure, examples of which are illustrated in the accompanying drawings.

(10) FIG. 1 is a schematic circuit diagram illustrating an organic light emitting display device of the present disclosure.

(11) As illustrated in FIG. 1, a gate line GL and a data line DL, which cross each other to define a pixel (pixel region) P, and a power line PL are formed in an organic light display device. A switching thin film transistor (TFT) Ts, a driving TFT Td, a storage capacitor Cst and an OLED D are formed in the pixel region P. The pixel region P may include a red pixel, a green pixel and a blue pixel.

(12) The switching thin film transistor Ts is connected to the gate line GL and the data line DL, and the driving thin film transistor Td and the storage capacitor Cst are connected between the switching thin film transistor Ts and the power line PL. The OLED D is connected to the driving thin film transistor Td. When the switching thin film transistor Ts is turned on by the gate signal applied through the gate line GL, the data signal applied through the data line DL is applied a gate electrode of the driving thin film transistor Td and one electrode of the storage capacitor Cst through the switching thin film transistor Ts.

(13) The driving thin film transistor Td is turned on by the data signal applied into the gate electrode so that a current proportional to the data signal is supplied from the power line PL to the OLED D through the driving thin film transistor Tr. The OLED D emits light having a luminance proportional to the current flowing through the driving thin film transistor Td. In this case, the storage capacitor Cst is charge with a voltage proportional to the data signal so that the voltage of

the gate electrode in the driving thin film transistor Td is kept constant during one frame.

Therefore, the organic light emitting display device can display a desired image.

(14) FIG. 2 is a schematic cross-sectional view illustrating an organic light emitting display device according to a first embodiment of the present disclosure.

(15) As illustrated in FIG. 2, the organic light emitting display device **100** includes a substrate **110**, a TFT Tr and an OLED D connected to the TFT Tr. For example, the organic light emitting display device **100** may include a red pixel, a green pixel and a blue pixel, and the OLED D may be formed in each of the red, green and blue pixels. Namely, the OLEDs D emitting red light, green light and blue light may be provided in the red, green and blue pixels, respectively.

(16) The substrate **110** may be a glass substrate or a plastic substrate. For example, the substrate **110** may be a polyimide substrate.

(17) A buffer layer **120** is formed on the substrate, and the TFT Tr is formed on the buffer layer **120**. The buffer layer **120** may be omitted.

(18) A semiconductor layer **122** is formed on the buffer layer **120**. The semiconductor layer **122** may include an oxide semiconductor material or polycrystalline silicon.

(19) When the semiconductor layer **122** includes the oxide semiconductor material, a light-shielding pattern (not shown) may be formed under the semiconductor layer **122**. The light to the semiconductor layer **122** is shielded or blocked by the light-shielding pattern such that thermal degradation of the semiconductor layer **122** can be prevented. On the other hand, when the semiconductor layer **122** includes polycrystalline silicon, impurities may be doped into both sides of the semiconductor layer **122**.

(20) A gate insulating layer **124** is formed on the semiconductor layer **122**. The gate insulating layer **124** may be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

(21) A gate electrode **130**, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer **124** to correspond to a center of the semiconductor layer **122**.

(22) In FIG. 2, the gate insulating layer **124** is formed on an entire surface of the substrate **110**. Alternatively, the gate insulating layer **124** may be patterned to have the same shape as the gate electrode **130**.

(23) An interlayer insulating layer **132**, which is formed of an insulating material, is formed on the gate electrode **130**. The interlayer insulating layer **132** may be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

(24) The interlayer insulating layer **132** includes first and second contact holes **134** and **136** exposing both sides of the semiconductor layer **122**. The first and second contact holes **134** and **136** are positioned at both sides of the gate electrode **130** to be spaced apart from the gate electrode **130**.

(25) The first and second contact holes **134** and **136** are formed through the gate insulating layer **124**. Alternatively, when the gate insulating layer **124** is patterned to have the same shape as the gate electrode **130**, the first and second contact holes **134** and **136** is formed only through the interlayer insulating layer **132**.

(26) A source electrode **140** and a drain electrode **142**, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer **132**.

(27) The source electrode **140** and the drain electrode **142** are spaced apart from each other with respect to the gate electrode **130** and respectively contact both sides of the semiconductor layer **122** through the first and second contact holes **134** and **136**.

(28) The semiconductor layer **122**, the gate electrode **130**, the source electrode **140** and the drain electrode **142** constitute the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr may correspond to the driving TFT Td (of FIG. 1).

(29) In the TFT Tr, the gate electrode **130**, the source electrode **140**, and the drain electrode **142** are positioned over the semiconductor layer **122**. Namely, the TFT Tr has a coplanar structure.

(30) Alternatively, in the TFT Tr, the gate electrode may be positioned under the semiconductor layer, and the source and drain electrodes may be positioned over the semiconductor layer such that the TFT Tr may have an inverted staggered structure. In this instance, the semiconductor layer may include amorphous silicon.

(31) Although not shown, the gate line and the data line cross each other to define the pixel, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element.

(32) In addition, the power line, which may be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame may be further formed.

(33) A passivation layer **150**, which includes a drain contact hole **152** exposing the drain electrode **142** of the TFT Tr, is formed to cover the TFT Tr.

(34) A first electrode **160**, which is connected to the drain electrode **142** of the TFT Tr through the drain contact hole **152**, is separately formed in each pixel. The first electrode **160** may be an anode and may be formed of a conductive material having a relatively high work function. For example, the first electrode **160** may be formed of a transparent conductive material such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO).

(35) When the OLED device **100** is operated in a top-emission type, a reflection electrode or a reflection layer may be formed under the first electrode **160**. For example, the reflection electrode or the reflection layer may be formed of aluminum-palladium-copper (APC) alloy.

(36) A bank layer **166** is formed on the passivation layer **150** to cover an edge of the first electrode **160**. Namely, the bank layer **166** is positioned at a boundary of the pixel and exposes a center of the first electrode **160** in the pixel.

(37) An organic emitting layer **162** is formed on the first electrode **160**. The organic emitting layer **162** may have a single-layered structure of an emitting material layer including an emitting material. To increase an emitting efficiency of the OLED D and/or the organic light emitting display device **100**, the organic emitting layer **162** may have a multi-layered structure. For example, the organic emitting layer **162** may include the EML, an electron blocking layer (EBL) between the first electrode **160** and the EML, and a hole blocking layer (HBL) between the EML and the second electrode **164**.

(38) The organic emitting layer **162** is separated in each of the red, green and blue pixels. As illustrated below, the organic emitting layer **162** in the blue pixel includes a first host of an anthracene derivative (compound) and a second host of a deuterated anthracene derivative such that the emitting efficiency and the lifespan of the OLED D in the blue pixel are improved.

(39) In addition, the EBL includes an amine derivative as an electron blocking material, and the HBL includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED D and an organic light emitting device **100** is further improved.

(40) A second electrode **164** is formed over the substrate **110** where the organic emitting layer **162** is formed. The second electrode **164** covers an entire surface of the display area and may be formed of a conductive material having a relatively low work function to serve as a cathode. For example, the second electrode **164** may be formed of aluminum (Al), magnesium (Mg) or Al—Mg alloy.

(41) The first electrode **160**, the organic emitting layer **162** and the second electrode **164** constitute the OLED D.

(42) An encapsulation film **170** is formed on the second electrode **164** to prevent penetration of moisture into the OLED D. The encapsulation film **170** includes a first inorganic insulating layer **172**, an organic insulating layer **174** and a second inorganic insulating layer **176** sequentially stacked, but it is not limited thereto. The encapsulation film **170** may be omitted.

(43) A polarization plate (not shown) for reducing an ambient light reflection may be disposed over the top-emission type OLED D. For example, the polarization plate may be a circular polarization

plate.

(44) In addition, a cover window (not shown) may be attached to the encapsulation film **170** or the polarization plate. In this instance, the substrate **110** and the cover window have a flexible property such that a flexible display device may be provided.

(45) FIG. 3 is a schematic cross-sectional view illustrating an OLED having a single emitting unit for the organic light emitting display device according to the first embodiment of the present disclosure.

(46) As illustrated in FIG. 3, the OLED D includes the first and second electrodes **160** and **164**, which face each other, and the organic emitting layer **162** therebetween. The organic emitting layer **162** includes an EML **240** between the first and second electrodes **160** and **164**, an EBL **230** between the first electrode **160** and the EML **240**, and an HBL **250** between the EML **240** and the second electrode **164**.

(47) The first electrode **160** may be formed of a conductive material having a relatively high work function to serve as an anode. The second electrode **164** may be formed of a conductive material having a relatively low work function to serve as a cathode.

(48) The organic emitting layer **162** may further include a hole transporting layer (HTL) **220** between the first electrode **160** and the EBL **230**.

(49) Moreover, the organic emitting layer **162** may further include a hole injection layer (HIL) **210** between the first electrode **160** and the HTL **220** and an electron injection layer (EIL) **260** between the second electrode **164** and the HBL **250**.

(50) The EML **240** includes a first host **242** of an anthracene derivative, a second host **244** of a deuterated anthracene derivative and a blue dopant (not shown) and provides blue emission.

(51) The compound of the first host **242** may be represented by Formula 1:

(52) ##STR00001##

(53) In Formula 1, each of R.sub.1 and R.sub.2 is independently C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 heteroaryl group, and each of L.sub.1 and L.sub.2 is independently C.sub.6~C.sub.30 arylene group. Each of a and b is an integer of 0 or 1, and at least one of a and b is 0.

(54) For example, R.sub.1 may be phenyl or naphthyl, R.sub.2 may be naphthyl, dibenzofuranyl or fused dibenzofuranyl. Each of L.sub.1 and L.sub.2 may independently be phenylene.

(55) In an exemplary embodiment, the first host **242** may be a compound being one of the followings in Formula 2:

(56) ##STR00002## ##STR00003## ##STR00004## ##STR00005## ##STR00006##

##STR00007## ##STR00008## ##STR00009## ##STR00010## ##STR00011## ##STR00012##

(57) The second host **244** may be a deuterated compound of the first host **242**. Namely, a hydrogen atom of the first host **242** may be substituted by a deuterium atom to form the second host **244**. A part or all of the hydrogen atoms of the compound of the first host **242** may be substituted by the deuterium atom.

(58) For example, the compound of the second host **244** may be represented by Formula 3:

(59) ##STR00013##

(60) In Formula 3, the definition of R.sub.1, R.sub.2, L.sub.1, L.sub.2, a and b is same as Formula 1. In Formula 3, Dx, Dy, Dm and Dn denote a number of the deuterium atom, and each of x, y, m and n is independently a positive integer. For example, a summation of x, y, m and n may be 15 to 29.

(61) In an exemplary embodiment, the second host **244** of Formula 3 may be a compound being one of the followings in Formula 4:

(62) ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018##

##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##

(63) A compound of the blue dopant may be represented by Formula 5-1 or Formula 5-2, but it is not limited thereto.

(64) ##STR00024##

(65) In Formula 5-1, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3. Each of R.sub.11 and R.sub.12 is independently selected from the group consisting of C.sub.1~C.sub.20 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.6~C.sub.30 aryl amino group, or adjacent two among R.sub.11 or adjacent two among R.sub.12 form a fused aromatic ring or a hetero-aromatic ring. R.sub.13 is selected from the group consisting of C.sub.1~C.sub.10 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.5~C.sub.30 aromatic amino group. Each of X.sub.1 and X.sub.2 is independently oxygen (O) or NR.sub.14, and R.sub.14 is C.sub.6~C.sub.30 aryl group.

(66) ##STR00025##

(67) In Formula 5-2, each of Ar.sub.1, Ar.sub.2, Ar.sub.3 and Ar.sub.4 is independently selected from the group consisting of C.sub.6~C.sub.30 aryl group and C.sub.5~C.sub.30 hetero aryl group, and each of R.sub.1 and R.sub.2 is independently selected from the group consisting of hydrogen, C.sub.1~C.sub.20 alkyl group and C.sub.6~C.sub.30 aryl group. For example, each of Ar.sub.1, Ar.sub.2, Ar.sub.3 and Ar.sub.4 may be independently selected from the group consisting of phenyl, dibenzofuranyl, naphthyl and biphenyl and may be substituted by trifluoromethyl, cyano or fluorine (F). Each of R.sub.1 and R.sub.2 may be independently selected from the group consisting of hydrogen, isopropyl and phenyl.

(68) For example, the blue dopant in Formula 5-1 may be a compound being one of the followings in Formula 6-1:

(69) ##STR00026## ##STR00027## ##STR00028## ##STR00029## ##STR00030##

##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035## ##STR00036##
##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042##
##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048##
##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053## ##STR00054##
##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060##
##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066##
##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072##
##STR00073## ##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078##
##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083## ##STR00084##
##STR00085##

(70) For example, the blue dopant of Formula 5-2 may be a compound being one of the followings in Formula 6-2:

(71) ##STR00086## ##STR00087## ##STR00088## ##STR00089## ##STR00090##

##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096##
##STR00097## ##STR00098## ##STR00099## ##STR00100##

(72) The EBL **230** includes an amine derivative as an electron blocking material. The material of the EBL **230** may be represented by Formula 7:

(73) ##STR00101##

(74) In Formula 7, each of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is polycyclic aryl group. For example, two of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 may be polycyclic aryl group. The monocyclic aryl group may be phenyl, and the polycyclic aryl group may be a fused-aryl group. The polycyclic aryl group may be an aryl group in which at least two phenyl groups are fused.

(75) The electron blocking material of Formula 7 may be one of the followings of Formula 8:

(76) ##STR00102## ##STR00103## ##STR00104## ##STR00105## ##STR00106##
##STR00107## ##STR00108##

(77) The HBL **250** may include an azine derivative as a hole blocking material. For example, the material of the HBL **250** may be represented by Formula 9:

(78) ##STR00109##

(79) In Formula 9, each of Y.sub.1 to Y.sub.5 are independently CR.sub.1 or N, and one to three of Y.sub.1 to Y.sub.5 is N. R.sub.1 is independently hydrogen or C.sub.6~C.sub.30 aryl group. L is C.sub.6~C.sub.30 arylene group, and R.sub.2 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group. R.sub.3 is hydrogen, or adjacent two of R3 form a fused ring. "a" is 0 or 1, "b" is 1 or 2, and "c" is an integer of 0 to 4.

(80) The hole blocking material of Formula 9 may be one of the followings of Formula 10:

(81) ##STR00110## ##STR00111## ##STR00112## ##STR00113## ##STR00114##

##STR00115## ##STR00116##

(82) Alternatively, the HBL **250** may include a benzimidazole derivative as a hole blocking material. For example, the material of the HBL **250** may be represented by Formula 11:

(83) ##STR00117##

(84) In Formula 11, Ar is C.sub.10~C.sub.30 arylene group, R.sub.1 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group, and R.sub.2 is C.sub.1~C.sub.10 alkyl group or C.sub.6~C.sub.30 aryl group.

(85) For example, Ar may be naphthylene or anthracenylene, R.sub.1 may be benzimidazole or phenyl, and R.sub.2 may be methyl, ethyl or phenyl.

(86) The hole blocking material of Formula 11 may be one of the followings of Formula 12:

(87) ##STR00118##

(88) The HBL **250** may include one of the hole blocking material of Formula 9 and the hole blocking material of Formula 11.

(89) In this instance, a thickness of the EML **240** may be greater than each of a thickness of the EBL **230** and a thickness of the HBL **250** and may be smaller than a thickness of the HTL **220**. For example, the EML may have a thickness of about 150 to 250 Å, and each of the EBL **230** and the HBL **250** may have a thickness of about 50 to 150 Å. The HTL **220** may have a thickness of about 900 to 1100 Å. The EBL **230** and the HBL **250** may have the same thickness.

(90) The HBL **250** may include both the hole blocking material of Formula 9 and the hole blocking material of Formula 11. For example, in the HBL **250**, hole blocking material of Formula 9 and the hole blocking material of Formula 11 may have the same weight %.

(91) In this instance, a thickness of the EML **240** may be greater than a thickness of the EBL **230** and may be smaller than a thickness of the HBL **250**. In addition, the thickness of HBL **250** may be smaller than a thickness of the HTL **220**. For example, the EML may have a thickness of about 200 to 300 Å, and the EBL **230** may have a thickness of about 50 to 150 Å. The HBL **250** may have a thickness of about 250 to 350 Å, and the HTL **220** may have a thickness of about 800 to 1000 Å.

(92) The hole blocking material of Formula 9 and/or the hole blocking material of Formula 11 have an electron transporting property such that an electron transporting layer may be omitted. As a result, the HBL **250** directly contacts the EIL **260** or the second electrode **164** without the EIL **260**.

(93) In the OLED D of the present disclosure, a weight % ratio of the first host **242** to the second host **244** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **242** to the second host **244** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **242** to the second host **244** may be about 7:3. The OLED D and the organic light emitting display device of the present disclosure have advantages in the emitting efficiency and the lifespan.

(94) In addition, when the EML **240** includes the blue dopant of Formula 5-1, an image with narrow full width at half maximum (FWHM) and high color purity is provided.

(95) Moreover, the EBL **230** includes an amine derivative as an electron blocking material, and the HBL **250** includes at least one of an azine derivative and a benzimidazole derivative as a hole blocking material. Accordingly, the lifespan of the OLED D and an organic light emitting device **100** is further improved.

Synthesis of the First Host

1. Synthesis of the Compound Host1

(96) ##STR00119##

(97) 10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(naphthalen-1-yl)-1,3,2-dioxaborolane (1.45 g, 5.74 mmol), tris (dibenzylideneacetone)dipalladium(0) (Pd.sub.2(dba).sub.3) (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC (high-performance liquid chromatography). The reaction flask was cooled down to the room temperature, and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM), and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host1 of white powder (2.00 g, yield: 89%).

2. Synthesis of the Compound Host2

(98) ##STR00120##

(99) 10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(4-(naphthalen-4-yl)phenyl)-1,3,2-dioxaborolane (1.90 g, 5.74 mmol), Pd.sub.2(dba).sub.3) (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM) twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host2 of white powder (2.28 g, yield: 86%).

3. Synthesis of the Compound Host3

(100) ##STR00121##

(101) 10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(dibenzofuran-1-yl)-1,3,2-dioxaborolane (1.69 g, 5.74 mmol), Pd.sub.2(dba).sub.3) (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM) twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host3 of white powder (1.91 g, yield: 78%).

4. Synthesis of the Compound Host4

(102) ##STR00122##

(103) 10-bromo-9-(naphthalene-3-yl)-anthracene (2.00 g, 5.23 mmol), 4,4,5,5-tetramethyl-2-(dibenzofuran-1-yl)phenyl-1,3,2-dioxaborolane (2.12 g, 5.74 mmol), Pd.sub.2(dba).sub.3) (0.24 g, 0.26 mmol) and toluene (50 mL) were added into the flask (250 mL) in the dry box. The reaction flask was removed from the dry box, and sodium carbonate anhydride (20 mL, 2M) was added into the flask. The reactants were stirred and heated at 90° C. overnight with monitoring the reaction by HPLC. The reaction flask was cooled down to the room temperature and then an organic layer was separated from an aqueous layer. The aqueous layer was washed with dichloromethane (DCM)

twice and the organic layer was concentrated with the rotary vaporizer to obtain a gray powder. The gray power was purified with alumina, precipitated with hexane and performed column chromatography using silica gel to obtain the compound Host4 of white powder (2.31 g, yield: 82%).

Synthesis of the Second Host

1. Synthesis of the Compound Host32

(104) ##STR00123##

(105) Under N.sub.2 condition, AlCl.sub.3 (0.48 g, 3.6 mmol) was added to perdeuterobenzene solution (100 mL) in which the compound Host2 (5 g, 9.87 mmol) was dissolved. The mixture was stirred under the room temperature for 6 hrs, and D.sub.2O (50 mL) was added. After separating the aqueous layer and the organic layer, the aqueous layer was washed with CH.sub.2Cl.sub.2 (30 mL). The obtained organic layer was dried using magnesium sulfate, and the volatiles were removed by rotary evaporation. The crude product was purified by column chromatography to obtain the compound Host32 (4.5 g) of white powder.

2. Synthesis of the Compound Host34

(106) ##STR00124##

(107) Under N.sub.2 condition, AlCl.sub.3 (0.48 g, 3.6 mmol) was added to perdeuterobenzene solution (100 mL) in which the compound Host4 (5 g, 9.15 mmol) was dissolved. The mixture was stirred under the room temperature for 6 hrs, and D.sub.2O (50 mL) was added. After separating the aqueous layer and the organic layer, the aqueous layer was washed with CH.sub.2Cl.sub.2 (30 mL). The obtained organic layer was dried using magnesium sulfate, and the volatiles were removed by rotary evaporation. The crude product was purified by column chromatography to obtain the compound Host34 (4.8 g) of white powder.

Synthesis of the Blue Dopant

1. Synthesis of the Compound Dopant56

(1) 3-nitro-N, N-diphenylaniline

(108) ##STR00125##

(109) Under N.sub.2 condition, the flask containing 3-nitroaniline (25.0 g), iodide benzene (81.0 g), copper iodide (3.5 g), potassium carbonate (100.0 g) and orthodichlorobenzene (250 mL) was heated and stirred for 14 hours. The reaction solution was cooled to the room temperature, and ammonia water was added for liquid separation. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=3/7 (volume ratio)) to obtain 3-nitro-N, N-diphenylaniline (44.0 g).

(2) N1,N1-diphenylbenzene-1,3-diamine

(110) ##STR00126##

(111) Under N.sub.2 condition, acetic acid cooled in the ice-bath was added and stirred. 3-nitro-N,N-diphenylaniline (44.0 g) was dropped into the solution to avoid significant increase the reaction temperature. After the addition was completed, the mixture was stirred at the room temperature for 30 minutes, and the disappearance of the starting materials was checked. After the reaction was completed, a supernatant liquid was collected by decantation, neutralized with sodium carbonate, and extracted with ethyl acetate. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=9/1 (volume ratio)). After removing the solvent by distillation under reduced pressure, heptane was added and reprecipitated to obtain N1,N1-diphenylbenzene-1,3-diamine (36.0 g).

(3) N1,N1,N3-triphenylbenzene-1,3-diamine

(112) ##STR00127##

(113) Under N.sub.2 condition, the flask containing N1,N1-diphenylbenzene-1,3-diamine (60.0 g), Pd-132 (1.3 g), NaOtBu (33.5 g) and xylene (300 mL) was stirred by heating at 120° C. The solution of xylene (50 ml), in which bromobenzene (36.2 g) was dissolved, was slowly dropped to the solution, and the mixture was heated and stirred for 1 hour after completion of dropping. After

cooling the mixture to the room temperature, water and ethyl acetate were added for liquid separation. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=5/5 (volume ratio)) to obtain N1,N1,N3-triphenylbenzene-1,3-diamine (73.0 g).

(4) N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N3,N3-triphenylbenzene-1,3-diamine)

(114) ##STR00128##

(115) Under N.sub.2 condition, the flask containing N1,N1,N3-triphenylbenzene-1,3-diamine (20.0 g), 1-bromo-2,3-dichlorobenzene (6.4 g), Pd-132 (0.2 g), NaOtBu (sodium tert-butoxide, 6.8 g) and xylene (70 mL) was heated and stirred at 120° C. for 2 hrs. After cooling the mixture to the room temperature, water and ethyl acetate were added for liquid separation. The resultant was purified by silica gel column chromatography (developing solution: toluene/heptane=4/6 (volume ratio)) to obtain N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N3,N3-triphenylbenzene-1,3-diamine) (15.0 g).

(5) Dopant56

(116) ##STR00129##

(117) 1.7 M tert-butyllithium pentane solution (18.1 ml) was added into the flask containing N1,N1'-(2-chloro-1,3-phenylene)bis(N1,N,N3-triphenylbenzene-1,3-diamine) (12.0 g) and tert-butylbenzene (100 mL) with cooling in the ice bath under N.sub.2 condition. After heating up to 60° C. and stirring for 2 hrs, the component having boiling point lower than tert-butylbenzene was distilled off under reduced pressure. The mixture was cooled to -50° C., and boron tribromide (2.9 mL) was added. The mixture was heated up to the room temperature and stirred for 0.5 hour. The mixture was cooled again in the ice bath, and N,N-diisopropylethylamine (5.4 mL) was added. The mixture was stirred at the room temperature until the exotherm was finished. The mixture was heated to the temperature of 120° C. and stirred for 3 hrs. The solution was cooled to the room temperature, and an aqueous sodium acetate solution, which was cooled in the ice bath, and ethyl acetate was sequentially added to the solution. The insoluble solid was filtered and phase-separated. Then, the residue was purified by silica gel column chromatography (developing solution: toluene/heptane=5/5 (volume ratio)). The mixture was washed with heated heptane and ethyl acetate, and then reprecipitated with a mixed solvent of toluene and ethyl acetate to obtain Dopant56 (2.0 g).

2. Synthesis of the Compound Dopant167

(1) 3,3'-((2-bromo-1,3-phenylene)bis(oxy)) di-1,1'-biphenyl

(118) ##STR00130##

(119) Under N.sub.2 condition, the flask containing 2-bromo-1,3-difluorobenzene (12.0 g), [1,1'-biphenyl]-3-ol (23.0 g), potassium carbonate (34.0 g) and NMP (130 mL) was heated and stirred at 170° C. for 10 hrs. After the reaction was stopped, the reaction solution was cooled to the room temperature, and water and toluene were added for liquid separation. The solvent was distilled off under reduced pressure, and the resultant was purified by silica gel column chromatography (developing solution: heptane/toluene=7/3 (volume ratio)) to obtain 3,3'-((2-bromo-1,3-phenylene)bis(oxy)) di-1,1'-biphenyl (26.8 g) was obtained.

(2) Dopant167

(120) ##STR00131##

(121) Under N.sub.2 condition, the flask containing 3,3'-((2-bromo-1,3-phenylene)bis(oxy))di-1,1'-biphenyl (14.0 g) and xylene (100 mL) was cooled to -40° C., and n-butyllithium hexane solution (2.6 M, 11.5 mL) was dropped. After heating the mixture to the room temperature, the mixture was cooled again to -40° C., and boron tribromide (3.3 mL) was added thereto. The mixture was heated to the room temperature and stirred for 13 hrs. The mixture was cooled to 0° C., and N,N-diisopropylethylamine (9.7 ml) was added. The mixture was heated and stirred at 130° C. for 5 hrs. The reaction solution was cooled to the room temperature, and an aqueous sodium acetate solution cooled in the ice bath was added and stirred. The precipitated solid was collected by suction filtration. The obtained solid was washed with water, methanol and heptane in that order

and recrystallized from chlorobenzene to obtain the compound Dopant167 (8.9 g).

(122) [Organic Light Emitting Diode]

(123) The anode (ITO, 50 Å), the HIL (Formula 13 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 13, 1000 Å), the EBL (Formula 15, 100 Å), the EML (host (98 wt %) and dopant (2 wt %), 200 Å), the HBL (Formula 16, 100 Å), the EIL (Formula 17 (98 wt %) and Li (2 wt %), 200 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

(124) ##STR00132## ##STR00133##

1. COMPARATIVE EXAMPLES

(1) Comparative Example 1 (Ref1)

(125) The compound "Dopant56" in Formula 6-1 is used as the dopant, and the compound "Host2" is used as the host.

(2) Comparative Example 2 (Ref2)

(126) The compound "Dopant1" in Formula 6-2 is used instead of the compound "Dopant56" in Formula 6-1 of Comparative Example 1.

(3) Comparative Example 3 (Ref3)

(127) The compound "Host4" is used instead of the compound "Host2" of Comparative Example 1.

(4) Comparative Example 4 (Ref4)

(128) The compound "Host4" is used instead of the compound "Host2" of Comparative Example 2.

(5) Comparative Example 5 (Ref5)

(129) The compound "Host32" is used instead of the compound "Host2" of Comparative Example 1.

(6) Comparative Example 6 (Ref6)

(130) The compound "Host32" is used instead of the compound "Host2" of Comparative Example 2.

(7) Comparative Example 7 (Ref7)

(131) The compound "Host34" is used instead of the compound "Host2" of Comparative Example 1.

(8) Comparative Example 8 (Ref8)

(132) The compound "Host34" is used instead of the compound "Host2" of Comparative Example 2.

2. EXAMPLES

(1) Example 1 (Ex1)

(133) The compound "Dopant56" in Formula 6-1 is used as the dopant, and the compound "Host2" and the compound "Host32" are used as the host. ("Host2": "Host32"=9:1 (wt % ratio))

(2) Example 2 (Ex2)

(134) The weight % ratio of "Host2" to "Host32" is changed into 7:3 from Example 1. ("Host2": "Host32"=7:3 (wt % ratio))

(3) Example 3 (Ex3)

(135) The weight % ratio of "Host2" to "Host32" is changed into 5:5 from Example 1. ("Host2": "Host32"=5:5 (wt % ratio))

(4) Example 4 (Ex4)

(136) The weight % ratio of "Host2" to "Host32" is changed into 3:7 from Example 1. ("Host2": "Host32"=3:7 (wt % ratio))

(5) Example 5 (Ex5)

(137) The weight % ratio of "Host2" to "Host32" is changed into 1:9 from Example 1. ("Host2": "Host32"=1:9 (wt % ratio))

(6) Examples 6 to 10 (Ex6 to Ex10)

(138) The compound "Dopant1" in Formula 6-2 is used instead of the compound "Dopant56" in Examples 1 to 5.

(7) Example 11 (Ex11)

(139) The compound “Dopant56” in Formula 6-1 is used as the dopant, and the compound “Host4” and the compound “Host34” are used as the host. (“Host4”:“Host34”=9:1 (wt % ratio))

(8) Example 12 (Ex12)

(140) The weight % ratio of “Host4” to “Host34” is changed into 7:3 from Example 11.

(“Host4”:“Host34”=7:3 (wt % ratio))

(9) Example 13 (Ex13)

(141) The weight % ratio of “Host4” to “Host34” is changed into 5:5 from Example 11.

(“Host4”:“Host34”=5:5 (wt % ratio))

(10) Example 14 (Ex14)

(142) The weight % ratio of “Host4” to “Host34” is changed into 3:7 from Example 11.

(“Host4”:“Host34”=3:7 (wt % ratio))

(11) Example 15 (Ex15)

(143) The weight % ratio of “Host4” to “Host34” is changed into 1:9 from Example 11.

(“Host4”:“Host34”=1:9 (wt % ratio))

(16) Examples 16 to 20 (Ex16 to Ex20)

(144) The compound “Dopant1” in Formula 6-2 is used instead of the compound “Dopant56” in Examples 11 to 15.

(145) The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 1 to 8 and Examples 1 to 20 are measured and listed in Tables 1 to 4.

(146) TABLE-US-00001 TABLE 1 Host ratio 1st 2nd CIE FWHM T95 host host V Cd/A x y [nm]
[hr] Ref1 10 0 3.77 4.78 0.1390 0.0612 26 153 Ex1 9 1 3.77 4.78 0.1390 0.0612 26 168 Ex2 7
3 3.77 4.78 0.1390 0.0611 26 184 Ex3 5 5 3.78 4.73 0.1391 0.0610 26 214 Ex4 3 7 3.78 4.68
0.1392 0.0610 26 275 Ex5 1 9 3.78 4.59 0.1393 0.0605 26 298 Ref5 0 10 3.78 4.55 0.1393
0.0605 26 302

(147) TABLE-US-00002 TABLE 2 Host ratio 1st 2nd CIE FWHM T95 host host V Cd/A x y [nm]
[hr] Ref2 10 0 3.77 6.91 0.1379 0.0974 40 221 Ex6 9 1 3.77 6.91 0.1379 0.0974 40 243 Ex7 7
3 3.77 6.91 0.1379 0.0973 40 265 Ex8 5 5 3.78 6.84 0.1380 0.0972 40 309 Ex9 3 7 3.78 6.77
0.1381 0.0972 40 398 Ex10 1 9 3.78 6.63 0.1382 0.0965 40 431 Ref6 0 10 3.78 6.56 0.1382
0.0965 40 435

(148) TABLE-US-00003 TABLE 3 Host ratio 1st 2nd CIE FWHM T95 host host V Cd/A x y [nm]
[hr] Ref3 10 0 3.92 4.98 0.1378 0.0632 26 141 Ex11 9 1 3.92 4.98 0.1378 0.0632 26 155 Ex12
7 3 3.92 4.98 0.1378 0.0631 26 169 Ex13 5 5 3.93 4.93 0.1379 0.0630 26 197 Ex14 3 7
3.93 4.88 0.1380 0.0630 26 253 Ex15 1 9 3.93 4.79 0.1381 0.0625 26 274 Ref7 0 10 3.93 4.75
0.1381 0.0625 26 278

(149) TABLE-US-00004 TABLE 4 Host ratio 1st 2nd CIE FWHM T95 host host V Cd/A x y [nm]
[hr] Ref4 10 0 3.92 7.11 0.1367 0.0994 40 203 Ex16 9 1 3.92 7.11 0.1367 0.0994 40 224 Ex17
7 3 3.92 7.11 0.1367 0.0993 40 244 Ex18 5 5 3.93 7.04 0.1368 0.0992 40 284 Ex19 3 7 3.93
6.97 0.1369 0.0992 40 366 Ex20 1 9 3.93 6.83 0.1370 0.0985 40 397 Ref8 0 10 3.93 6.76
0.1370 0.0985 40 400

(150) As shown in Tables 1 to 4, the lifespan of the OLED in Examples 1 to 20 using the first host, which is an anthracene derivative non-substituted with deuterium, and the second host, which is an anthracene derivative substituted with deuterium, is significantly increased in comparison to the OLED in Comparative Examples 1 to 4 using the first host without the second host.

(151) In addition, in comparison to the OLED in Comparative Examples 5 to 8 using the second host without the first host, the lifespan of the OLED in Examples 5, 10, 15 and 20 is slightly decreased, but the emitting efficiency of the OLED in Examples 5, 10, 15 and 20 is improved.

(152) Accordingly, in the OLED of the present disclosure, a weight % ratio of the first host to the second host may be about 1:9 to 7:3. To provide sufficient emitting efficiency and lifespan of the OLED and the organic light emitting display device, the weight % ratio of the first host to the

second host may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host to the second host may be about 7:3.

(153) In the OLED D of the present disclosure, the EML **240** includes anthracene derivative as the first host **242** and deuterated anthracene derivative as the second host **244** such that the OLED D and the organic light emitting display device **100** has advantages in the emitting efficiency and the lifespan.

(154) The anode (ITO, 50 Å), the HIL (Formula 18 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 18, 1000 Å), the EBL (100 Å), the EML (host (98 wt %) and dopant (2 wt %), 200 Å), the HBL (100 Å), the EIL (Formula 17 (98 wt %) and Li (2 wt %), 200 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

(155) ##STR00134##

1. COMPARATIVE EXAMPLES

(1) Comparative Example 9 (Ref9)

(156) The compound of Formula 19 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” in Formula 2 is used as the host, and the compound “E1” in Formula 10 is used to form the HBL.

(2) Comparative Example 10 (Ref10)

(157) The compound “H3” in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 9.

(3) Comparative Example 11 (Ref11)

(158) The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 9.

(4) Comparative Example 12 (Ref12)

(159) The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 10.

(5) Comparative Examples 13 to 16 (Ref13 to Ref16)

(160) The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Examples 9 to 12.

(6) Comparative Examples 17 to 20 (Ref17 to Ref20)

(161) The compound “Host4” in Formula 2 is used instead of the compound “Host2” in Comparative Examples 9 to 12.

(7) Comparative Examples 21 to 24 (Ref21 to Ref24)

(162) The compound “Host4” in Formula 2 is used instead of the compound “Host2” in Comparative Examples 13 to 16.

(163) ##STR00135##

2. EXAMPLES

(1) Example 21 (Ex21)

(164) The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” and the compound “Host32” are used as the host, and the compound “E1” in Formula 10 is used to form the HBL. (“Host2”:“Host32”=3:7 (wt % ratio))

(2) Example 22 (Ex22)

(165) The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 21.

(3) Example 23 (Ex23)

(166) The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 21.

(4) Example 24 (Ex24)

(167) The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 22.

(5) Example 25 (Ex25)

(168) The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host4” and the compound “Host34” are used as the host, and the compound “E1” in Formula 10 is used to form the HBL. (“Host4”:“Host34”=3:7 (wt % ratio))

(6) Example 26 (Ex26)

(169) The compound “F1” in Formula 12 is used instead of the compound “E1” in Formula 10 in Comparative Example 25.

(7) Example 27 (Ex27)

(170) The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 25.

(8) Example 28 (Ex28)

(171) The compound “Dopant167” in Formula 6-1 is used instead of the compound “Dopant56” in Comparative Example 26.

(172) The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 9 to 24 and Examples 21 to 28 are measured and listed in Tables 5 and 6.

(173) TABLE-US-00005 TABLE 5 Host ratio 1st 2nd CIE T95 host host V Cd/A x y [hr] Ref9 10 0 3.77 4.70 0.1390 0.0612 45 Ref10 10 0 3.68 4.75 0.139 0.0614 232 Ex21 3 7 3.68 4.65 0.1410 0.0564 418 Ref11 10 0 3.47 5.00 0.1420 0.0562 20 Ref12 10 0 3.38 5.05 0.1420 0.0564 104 Ex22 3 7 3.38 4.95 0.1440 0.0514 188 Ref13 10 0 3.92 4.90 0.1378 0.1212 41 Ref14 10 0 3.83 4.95 0.1378 0.1214 213 Ex23 3 7 3.83 4.85 0.1398 0.1164 383 Ref15 10 0 3.62 5.20 0.1408 0.1162 18 Ref16 10 0 3.53 5.25 0.1408 0.1164 96 Ex24 3 7 3.53 5.15 0.1428 0.1114 172

(174) TABLE-US-00006 TABLE 6 Host ratio 1st 2nd CIE T95 host host V Cd/A x y [hr] Ref17 10 0 3.94 4.62 0.1389 0.0609 41 Ref18 10 0 3.85 4.67 0.1389 0.0612 211 Ex25 3 7 3.85 4.57 0.1409 0.0562 380 Ref19 10 0 3.64 4.92 0.1419 0.0559 18 Ref20 10 0 3.55 4.97 0.1419 0.0562 95 Ex26 3 7 3.55 4.87 0.1439 0.0512 171 Ref21 10 0 4.09 4.82 0.1377 0.1209 38 Ref22 10 0 4.00 4.87 0.1377 0.1212 194 Ex27 3 7 4.00 4.77 0.1397 0.1162 349 Ref23 10 0 3.79 5.12 0.1407 0.1159 17 Ref24 10 0 3.70 5.17 0.1407 0.1162 87 Ex28 3 7 3.70 5.07 0.1427 0.1112 157

(175) As shown in Tables 5 and 6, in comparison to the OLED in Comparative Examples 9, 11, 13, 15, 17, 19, 21 and 23, the properties of the driving voltage, the emitting efficiency and the lifespan of the OLED in Comparative Examples 10, 12, 14, 16, 18, 20, 22 and 24, where the EBL includes the amine derivative of Formula 7, and the HBL includes the azine derivative of Formula 9 or the benzimidazole derivative of Formula 11, are improved.

(176) In addition, when the EML of the OLED includes the first host of Formula 1, which is an anthracene derivative, and the second host of Formula 3, which is an anthracene derivative substituted with deuterium, as Examples 25 to 28, the emitting efficiency and the lifespan are further improved.

(177) [Organic Light Emitting Diode]

(178) The anode (ITO, 50 Å), the HIL (Formula 18 (97 wt %) and Formula 14 (3 wt %), 100 Å), the HTL (Formula 18, 900 Å), the EBL (100 Å), the EML (host (98 wt %) and dopant (2 wt %), 250 Å), the HBL (300 Å), the EIL (LiF, 10 Å) and the cathode (Al, 500 Å) was sequentially deposited to form the OLED.

1. COMPARATIVE EXAMPLES

(1) Comparative Example 25 (Ref25)

(179) The compound of Formula 19 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” in Formula 2 is used as the host, and the compound “E16” in Formula 10 and the compound “F1” in Formula 12 are used to form the HBL. (“E16”:“F1”=1:1 (wt % ratio))

(2) Comparative Example 26 (Ref26)

(180) The compound of Formula 20 is used instead of the compound of Formula 19 in Comparative Example 25.

(3) Comparative Example 27 (Ref27)

(181) The compound “H3” in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 25.

(4) Comparative Example 28 (Ref28)

(182) The compound of Formula 19 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host4” in Formula 2 is used as the host, and the compound “E16” in Formula 10 and the compound “F1” in Formula 12 are used to form the HBL. (“E16”：“F1”=1:1 (wt % ratio))

(5) Comparative Example 29 (Ref29)

(183) The compound of Formula 20 is used instead of the compound of Formula 19 in Comparative Example 28.

(6) Comparative Example 30 (Ref30)

(184) The compound “H3” in Formula 8 is used instead of the compound of Formula 19 in Comparative Example 28.

(185) ##STR00136##

2. EXAMPLES

(1) Example 29

(186) The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host2” and the compound “Host32” are used as the host, and the compound “E16” in Formula 10 and the compound “F1” in Formula 12 are used to form the HBL. (“Host2”：“Host32”=3:7 (wt % ratio), “E16”：“F1”=1:1 (wt % ratio))

(2) Example 30

(187) The compound “H3” of Formula 8 is used to form the EBL, the compound “Dopant56” in Formula 6-1 is used as the dopant, the compound “Host4” and the compound “Host34” are used as the host, and the compound “E16” in Formula 10 and the compound “F1” in Formula 12 are used to form the HBL. (“Host4”：“Host34”=3:7 (wt % ratio), “E16”：“F1”=1:1 (wt % ratio))

(188) The properties, i.e., voltage (V), efficiency (Cd/A), color coordinate (CIE), FWHM and lifespan (T95), of the OLEDs manufactured in Comparative Examples 25 to 30 and Examples 29 and 30 are measured and listed in Table 7.

(189) TABLE-US-00007

TABLE 7	Host ratio	1st	2nd	CIE	T95	host	host	V	Cd/A	x	y	[hr]	Ref																																																		
25	10	0	3.67	4.94	0.1420	0.0600	44	Ref26	10	0	3.55	4.76	0.1362	0.0606	31	Ref27	10	0	3.58	4.99	0.1360	0.0602	227	Ex29	3	7	3.58	4.89	0.1380	0.0552	409	Ref28	10	0	3.84	4.86	0.1430	0.0630	40	Ref29	10	0	3.72	4.68	0.1352	0.0636	28	Ref30	10	0	3.75	4.91	0.1350	0.0632	207	Ex30	3	7	3.75	4.81	0.1370	0.0582	373

(190) As shown in Table 7, in comparison to the OLED in Comparative Examples 25, 26, 28 and 29, the emitting efficiency of the OLED in Comparative Examples 27 and 30 is improved, and the lifespan of the OLED in Comparative Examples 27 and 30 is significantly increased.

(191) In addition, when the EML of the OLED includes the first host of Formula 1, which is an anthracene derivative, and the second host of Formula 3, which is an anthracene derivative substituted with deuterium, as Examples 29 and 30, the emitting efficiency and the lifespan are further improved.

(192) FIG. 4 is a schematic cross-sectional view illustrating an OLED having a tandem structure of two emitting units according to the first embodiment of the present disclosure.

(193) As shown in FIG. 4, the OLED D includes the first and second electrodes **160** and **164** facing each other and the organic emitting layer **162** between the first and second electrodes **160** and **164**. The organic emitting layer **162** includes a first emitting part **310** including a first EML **320**, a second emitting part **330** including a second EML **340** and a charge generation layer (CGL) **350** between the first and second emitting parts **310** and **330**.

(194) The first electrode **160** may be formed of a conductive material having a relatively high work function to serve as an anode for injecting a hole into the organic emitting layer **162**. The second electrode **164** may be formed of a conductive material having a relatively low work function to serve as a cathode for injecting an electron into the organic emitting layer **162**.

(195) The CGL **350** is positioned between the first and second emitting parts **310** and **330**, and the first emitting part **310**, the CGL **350** and the second emitting part **330** are sequentially stacked on the first electrode **160**. Namely, the first emitting part **310** is positioned between the first electrode **160** and the CGL **350**, and the second emitting part **320** is positioned between the second electrode **160** and the CGL **350**.

(196) The first emitting part **310** includes a first EML **320**, a first EBL **316** between the first electrode **160** and the first EML **320** and a first HBL between the first EML **320** and the CGL **350**.

(197) In addition, the first emitting part **310** may further include a first HTL **314** between the first electrode **160** and the first EBL **316** and an HIL **312** between the first electrode **160** and the first HTL **314**.

(198) The first EML **320** includes a first host **322**, which is an anthracene derivative, a second host **324**, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the first EML **320**.

(199) Namely, the first EML **320** may include the compound of Formula 1 as the first host **322**, the compound of Formula 3 as the second host **324** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

(200) In the first EML **320**, a weight % ratio of the first host **322** to the second host **324** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **322** to the second host **324** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **322** to the second host **324** may be about 7:3.

(201) The first EBL **316** may include an electron blocking material of Formula 7. The first HBL **318** may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

(202) The second emitting part **330** includes the second EML **340**, a second EBL **334** between the CGL **350** and the second EML **340** and a second HBL **336** between the second EML **340** and the second electrode **164**.

(203) In addition, the second emitting part **330** may further include a second HTL **332** between the CGL **350** and the second EBL **334** and an EIL **338** between the second HBL **336** and the second electrode **164**.

(204) The second EML **340** includes a first host **342**, which is an anthracene derivative, a second host **344**, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the second EML **340**.

(205) Namely, the second EML **340** may include the compound of Formula 1 as the first host **342**, the compound of Formula 3 as the second host **344** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

(206) In the second EML **340**, a weight % ratio of the first host **342** to the second host **344** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **342** to the second host **344** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **342** to the second host **344** may be about 7:3.

(207) The first host **342** of the second EML **340** may be same as or different from the first host **322** of the first EML **320**, and the second host **344** of the second EML **340** may be same as or different from the second host **324** of the first EML **320**. In addition, the blue dopant of the second EML **340**

may be same as or different from the blue dopant of the first EML **320**.

(208) The second EBL **334** may include an electron blocking material of Formula 7. The second HBL **336** may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

(209) The CGL **350** is positioned between the first and second emitting parts **310** and **330**. Namely, the first and second emitting parts **310** and **330** are connected through the CGL **350**. The CGL **350** may be a P-N junction CGL of an N-type CGL **352** and a P-type CGL **354**.

(210) The N-type CGL **352** is positioned between the first HBL **318** and the second HTL **332**, and the P-type CGL **354** is positioned between the N-type CGL **352** and the second HTL **332**.

(211) In the OLED D, since each of the first and second EMLs **320** and **340** includes the first host **322** and **342**, each of which is an anthracene derivative, and the second host **324** and **344**, each of which is a deuterated anthracene derivative, the OLED D and the organic light emitting display device **100** have advantages in the emitting efficiency and the lifespan.

(212) In addition, at least one of the first and second EBLs **316** and **334** includes an amine derivative of Formula 7, and at least one of the first and second HBLs **318** and **336** includes at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11. As a result, the lifespan of the OLED D and the organic light emitting display device **100** is further improved.

(213) Moreover, since the first and second emitting parts **310** and **330** for emitting blue light are stacked, the organic light emitting display device **100** provides an image having high color temperature.

(214) FIG. 5 is a schematic cross-sectional view illustrating an organic light emitting display device according to a second embodiment of the present disclosure, and FIG. 6 is a schematic cross-sectional view illustrating an OLED for the organic light emitting display device according to the second embodiment of the present disclosure.

(215) As shown in FIG. 5, the organic light emitting display device **400** includes a first substrate **410**, where a red pixel BP, a green pixel GP and a blue pixel BP are defined, a second substrate **470** facing the first substrate **410**, an OLED D, which is positioned between the first and second substrates **410** and **470** and providing white emission, and a color filter layer **480** between the OLED D and the second substrate **470**.

(216) Each of the first and second substrates **410** and **470** may be a glass substrate or a plastic substrate. For example, each of the first and second substrates **410** and **470** may be a polyimide substrate.

(217) A buffer layer **420** is formed on the substrate, and the TFT Tr corresponding to each of the red, green and blue pixels RP, GP and BP is formed on the buffer layer **420**. The buffer layer **420** may be omitted.

(218) A semiconductor layer **422** is formed on the buffer layer **420**. The semiconductor layer **122** may include an oxide semiconductor material or polycrystalline silicon.

(219) A gate insulating layer **424** is formed on the semiconductor layer **422**. The gate insulating layer **424** may be formed of an inorganic insulating material such as silicon oxide or silicon nitride.

(220) A gate electrode **430**, which is formed of a conductive material, e.g., metal, is formed on the gate insulating layer **424** to correspond to a center of the semiconductor layer **422**.

(221) An interlayer insulating layer **432**, which is formed of an insulating material, is formed on the gate electrode **430**. The interlayer insulating layer **432** may be formed of an inorganic insulating material, e.g., silicon oxide or silicon nitride, or an organic insulating material, e.g., benzocyclobutene or photo-acryl.

(222) The interlayer insulating layer **432** includes first and second contact holes **434** and **436** exposing both sides of the semiconductor layer **422**. The first and second contact holes **434** and **436** are positioned at both sides of the gate electrode **430** to be spaced apart from the gate electrode **430**.

(223) A source electrode **440** and a drain electrode **442**, which are formed of a conductive material, e.g., metal, are formed on the interlayer insulating layer **432**.

(224) The source electrode **440** and the drain electrode **442** are spaced apart from each other with respect to the gate electrode **430** and respectively contact both sides of the semiconductor layer **422** through the first and second contact holes **434** and **436**.

(225) The semiconductor layer **422**, the gate electrode **430**, the source electrode **440** and the drain electrode **442** constitute the TFT Tr. The TFT Tr serves as a driving element. Namely, the TFT Tr may correspond to the driving TFT Td (of FIG. 1).

(226) Although not shown, the gate line and the data line cross each other to define the pixel, and the switching TFT is formed to be connected to the gate and data lines. The switching TFT is connected to the TFT Tr as the driving element.

(227) In addition, the power line, which may be formed to be parallel to and spaced apart from one of the gate and data lines, and the storage capacitor for maintaining the voltage of the gate electrode of the TFT Tr in one frame may be further formed.

(228) A passivation layer **450**, which includes a drain contact hole **452** exposing the drain electrode **442** of the TFT Tr, is formed to cover the TFT Tr.

(229) A first electrode **460**, which is connected to the drain electrode **442** of the TFT Tr through the drain contact hole **452**, is separately formed in each pixel. The first electrode **160** may be an anode and may be formed of a conductive material having a relatively high work function. For example, the first electrode **460** may be formed of a transparent conductive material such as indium-tin-oxide (ITO) or indium-zinc-oxide (IZO).

(230) A reflection electrode or a reflection layer may be formed under the first electrode **460**. For example, the reflection electrode or the reflection layer may be formed of aluminum-palladium-copper (APC) alloy.

(231) A bank layer **466** is formed on the passivation layer **450** to cover an edge of the first electrode **460**. Namely, the bank layer **466** is positioned at a boundary of the pixel and exposes a center of the first electrode **460** in the red, green and blue pixels RP, GP and BP. The bank layer **466** may be omitted.

(232) An organic emitting layer **462** is formed on the first electrode **460**.

(233) Referring to FIG. 6, the organic emitting layer **462** includes a first emitting part **530** including a first EML **520**, a second emitting part **550** including a second EML **540**, a third emitting part **570** including a third EML **560**, a first CGL **580** between the first and second emitting parts **530** and **550** and a second CGL **590** between the second and third emitting parts **550** and **570**.

(234) The first electrode **460** may be formed of a conductive material having a relatively high work function to serve as an anode for injecting a hole into the organic emitting layer **462**. The second electrode **464** may be formed of a conductive material having a relatively low work function to serve as a cathode for injecting an electron into the organic emitting layer **462**.

(235) The first CGL **580** is positioned between the first and second emitting parts **530** and **550**, and the second CGL **590** is positioned between the second and third emitting parts **550** and **570**. Namely, the first emitting part **530**, the first CGL **580**, the second emitting part **550**, the second CGL **590** and the third emitting part **570** are sequentially stacked on the first electrode **460**. In other words, the first emitting part **530** is positioned between the first electrode **460** and the first CGL **580**, the second emitting part **550** is positioned between the first and second CGLs **580** and **590**, and the third emitting part **570** is positioned between the second electrode **460** and the second CGL **590**.

(236) The first emitting part **530** may include an HIL **532**, a first HTL **534**, a first EBL **536**, the first EML **520** and a first HBL **538** sequentially stacked on the first electrode **460**. Namely, the HIL **532**, the first HTL **534** and the first EBL **536** are positioned between the first electrode **460** and the first EML **520**, and the first HBL **538** is positioned between the first EML **520** and the first CGL **580**.

(237) The first EML **520** includes a first host **522**, which is an anthracene derivative, a second host

524, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the first EML **520**.

(238) Namely, the first EML **520** may include the compound of Formula 1 as the first host **522**, the compound of Formula 3 as the second host **524** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

(239) In the first EML **520**, a weight % ratio of the first host **522** to the second host **524** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **522** to the second host **524** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **522** to the second host **524** may be about 7:3.

(240) The first EBL **536** may include an electron blocking material of Formula 7. The first HBL **538** may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11.

(241) The second EML **550** may include a second HTL **552**, the second EML **540** and an electron transporting layer (ETL) **554**. The second HTL **552** is positioned between the first CGL **580** and the second EML **540**, and the ETL **554** is positioned between the second EML **540** and the second CGL **590**.

(242) The second EML **540** may be a yellow-green EML. For example, the second EML **540** may include a host and a yellow-green dopant. Alternatively, the second EML **540** may include a host, a red dopant and a green dopant. In this instance, the second EML **540** may include a lower layer including the host and the red dopant (or the green dopant) and an upper layer including the host and the green dopant (or the red dopant).

(243) The third emitting part **570** may include a third HTL **572**, a second EBL **574**, the third EML **560**, a second HBL **576** and an EIL **578**. The third EML **560** (or the third emitting part **570**) includes a first host **562**, which is an anthracene derivative, a second host **564**, which is a deuterated anthracene derivative, and a blue dopant (not shown) such that blue light is provided from the third EML **560**. Namely, the third EML **560** may include the compound of Formula 1 as the first host **562**, the compound of Formula 3 as the second host **564** and the compound of Formula 5-1 or Formula 5-2 as the blue dopant.

(244) In the third EML **560**, a weight % ratio of the first host **562** to the second host **564** may be about 1:9 to about 9:1, preferably about 1:9 to about 7:3. To provide sufficient emitting efficiency and lifespan of the OLED D and the organic light emitting display device, the weight % ratio of the first host **562** to the second host **564** may be about 3:7. On the other hand, to increase the lifespan without decrease of the emitting efficiency, the weight % ratio of the first host **562** to the second host **564** may be about 7:3.

(245) The first host **562** of the third EML **560** may be same as or different from the first host **522** of the first EML **520**, and the second host **564** of the third EML **560** may be same as or different from the second host **524** of the first EML **520**. In addition, the blue dopant of the third EML **560** may be same as or different from the blue dopant of the first EML **520**.

(246) The second EBL **574** may include an electron blocking material of Formula 7. The second HBL **576** may include at least one of a hole blocking material of Formula 9 and a hole blocking material of Formula 11. The electron blocking material of the second EBL **574** may be same as or different from the electron blocking material of the first EBL **536**, and the hole blocking material of the second HBL **576** may be same as or different from the hole blocking material of the first HBL **538**.

(247) The first CGL **580** is positioned between the first emitting part **530** and the second emitting part **550**, and the second CGL **590** is positioned between the second emitting part **550** and the third emitting part **570**. Namely, the first and second emitting stacks **530** and **550** are connected through the first CGL **580**, and the second and third emitting stacks **550** and **570** are connected through the

second CGL **590**. The first CGL **580** may be a P-N junction CGL of a first N-type CGL **582** and a first P-type CGL **584**, and the second CGL **590** may be a P-N junction CGL of a second N-type CGL **592** and a second P-type CGL **594**.

(248) In the first CGL **580**, the first N-type CGL **582** is positioned between the first HBL **538** and the second HTL **552**, and the first P-type CGL **584** is positioned between the first N-type CGL **582** and the second HTL **552**.

(249) In the second CGL **590**, the second N-type CGL **592** is positioned between the ETL **554** and the third HTL **572**, and the second P-type CGL **594** is positioned between the second N-type CGL **592** and the third HTL **572**.

(250) In the OLED D, since each of the first and third EMLs **520** and **560** includes the first host **522** and **562**, each of which is an anthracene derivative, the second host **524** and **564**, each of which is a deuterated anthracene derivative, and the blue dopant.

(251) Accordingly, the OLED D including the first and third emitting parts **530** and **570** with the second emitting part **550**, which emits yellow-green light or red/green light, can emit white light.

(252) In FIG. **6**, the OLED D has a triple-stack structure of the first, second and third emitting parts **530**, **550** and **570**. Alternatively, the OLED D may have a double-stack structure without the first emitting part **530** and the third emitting part **570**.

(253) Referring to FIG. **5** again, a second electrode **464** is formed over the substrate **110** where the organic emitting layer **162** is formed.

(254) In the organic light emitting display device **400**, since the light emitted from the organic emitting layer **462** is incident to the color filter layer **480** through the second electrode **464**, the second electrode **464** has a thin profile for transmitting the light.

(255) The first electrode **460**, the organic emitting layer **462** and the second electrode **464** constitute the OLED D.

(256) The color filter layer **480** is positioned over the OLED D and includes a red color filter **482**, a green color filter **484** and a blue color filter **486** respectively corresponding to the red, green and blue pixels RP, GP and BP.

(257) Although not shown, the color filter layer **480** may be attached to the OLED D by using an adhesive layer. Alternatively, the color filter layer **480** may be formed directly on the OLED D.

(258) An encapsulation film (not shown) may be formed to prevent penetration of moisture into the OLED D. For example, the encapsulation film may include a first inorganic insulating layer, an organic insulating layer and a second inorganic insulating layer sequentially stacked, but it is not limited thereto. The encapsulation film may be omitted.

(259) A polarization plate (not shown) for reducing an ambient light reflection may be disposed over the top-emission type OLED D. For example, the polarization plate may be a circular polarization plate.

(260) In FIG. **5**, the light from the OLED D passes through the second electrode **464**, and the color filter layer **480** is disposed on or over the OLED D. Alternatively, when the light from the OLED D passes through the first electrode **460**, the color filter layer **480** may be disposed between the OLED D and the first substrate **410**.

(261) A color conversion layer (not shown) may be formed between the OLED D and the color filter layer **480**. The color conversion layer may include a red color conversion layer, a green color conversion layer and a blue color conversion layer respectively corresponding to the red, green and blue pixels RP, GP and BP. The white light from the OLED D is converted into the red light, the green light and the blue light by the red, green and blue color conversion layer, respectively.

(262) As described above, the white light from the organic light emitting diode D passes through the red color filter **482**, the green color filter **484** and the blue color filter **486** in the red pixel RP, the green pixel GP and the blue pixel BP such that the red light, the green light and the blue light are provided from the red pixel RP, the green pixel GP and the blue pixel BP, respectively.

(263) In FIGS. **5** and **6**, the OLED D emitting the white light is used for a display device.

Alternatively, the OLED D may be formed on an entire surface of a substrate without at least one of the driving element and the color filter layer to be used for a lightening device. The display device and the lightening device each including the OLED D of the present disclosure may be referred to as an organic light emitting device.

(264) FIG. 7 is a schematic cross-sectional view illustrating an organic light emitting display device according to a third embodiment of the present disclosure.

(265) As shown in FIG. 7, the organic light emitting display device **600** includes a first substrate **610**, where a red pixel BP, a green pixel GP and a blue pixel BP are defined, a second substrate **670** facing the first substrate **610**, an OLED D, which is positioned between the first and second substrates **610** and **670** and providing white emission, and a color conversion layer **680** between the OLED D and the second substrate **470**.

(266) Although not shown, a color filter may be formed between the second substrate **670** and each color conversion layer **680**.

(267) A TFT Tr, which corresponding to each of the red, green and blue pixels RP, GP and BP, is formed on the first substrate **610**, and a passivation layer **650**, which has a drain contact hole **652** exposing an electrode, e.g., a drain electrode, of the TFT Tr is formed to cover the TFT Tr.

(268) The OLED D including a first electrode **660**, an organic emitting layer **662** and a second electrode **664** is formed on the passivation layer **650**. In this instance, the first electrode **660** may be connected to the drain electrode of the TFT Tr through the drain contact hole **652**.

(269) An bank layer **666** covering an edge of the first electrode **660** is formed at a boundary of the red, green and blue pixel regions RP, GP and BP.

(270) The OLED D emits a blue light and may have a structure shown in FIG. 3 or FIG. 4. Namely, the OLED D is formed in each of the red, green and blue pixels RP, GP and BP and provides the blue light.

(271) The color conversion layer **680** includes a first color conversion layer **682** corresponding to the red pixel RP and a second color conversion layer **684** corresponding to the green pixel GP. For example, the color conversion layer **680** may include an inorganic color conversion material such as a quantum dot.

(272) The blue light from the OLED D is converted into the red light by the first color conversion layer **682** in the red pixel RP, and the blue light from the OLED D is converted into the green light by the second color conversion layer **684** in the green pixel GP.

(273) Accordingly, the organic light emitting display device **600** can display a full-color image.

(274) On the other hand, when the light from the OLED D passes through the first substrate **610**, the color conversion layer **680** is disposed between the OLED D and the first substrate **610**.

(275) While the present disclosure has been described with reference to exemplary embodiments and examples, these embodiments and examples are not intended to limit the scope of the present disclosure. Rather, it will be apparent to those skilled in the art that various modifications and variations can be made in the present disclosure without departing from the spirit or scope of the invention. Thus, it is intended that the present disclosure cover the modifications and variations of the present disclosure provided they come within the scope of the appended claims and their equivalents.

(276) The various embodiments described above can be combined to provide further embodiments. All of patents, patent application publications, patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

(277) These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be

construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. An organic light emitting diode (OLED), comprising: a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is a non-deuterated anthracene derivative, and the second host is a deuterated anthracene derivative, wherein a weight % ratio of the first host to the second host is 3:7 to 9:1, wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative, and wherein the second host has only difference of substitution of deuterium atom from the first host.
2. The OLED of claim 1, wherein the first host is represented by Formula 1: ##STR00137## wherein each of R.sub.1 and R.sub.2 is independently C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 heteroaryl group, and each of L.sub.1 and L.sub.2 is independently C.sub.6~C.sub.30 arylene group, and wherein each of a and b is an integer of 0 or 1, and at least one of a and b is 0.
3. The OLED of claim 2, wherein the first host is a compound being one of the followings of Formula 2: ##STR00138## ##STR00139## ##STR00140## ##STR00141## ##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148##
4. The OLED of claim 2, wherein the second host is represented by Formula 3: ##STR00149## wherein the definition of R.sub.1, R.sub.2, L.sub.1, L.sub.2, a and b is same as Formula 1, and wherein each of x, y, m and n is independently a positive integer, and a summation of x, y, m and n may be 15 to 29.
5. The OLED of claim 4, wherein the second host is a compound being one of the followings of Formula 12: ##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158## ##STR00159## ##STR00160##
6. The OLED of one of claim 1, wherein the second host is a compound in which all or part of hydrogen of the first host is substituted with deuterium.
7. The OLED of claim 1, wherein the weight % ratio of the first host to the second host is 3:7.
8. The OLED of claim 1, wherein the weight % ratio of the first host to the second host is 7:3.
9. The OLED of claim 1, wherein the blue dopant is represented by Formula 4 or Formula 5: ##STR00161## wherein in Formula 4, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3, wherein each of R.sub.11 and R.sub.12 is independently selected from the group consisting of C.sub.1~C.sub.20 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.6~C.sub.30 aryl amino group, or adjacent two among R.sub.11 or adjacent two among R.sub.12 form a fused aromatic ring or a hetero-aromatic ring, wherein R.sub.13 is selected from the group consisting of C.sub.1~C.sub.10 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.5~C.sub.30 aromatic amino group, wherein each of X.sub.1 and X.sub.2 is independently oxygen (O) or NR.sub.14, and R.sub.14 is C.sub.6~C.sub.30 aryl group, and wherein in Formula 5, each of Ar.sub.1, Ar.sub.2, Ar.sub.3 and Ar.sub.4 is independently selected from the group consisting of C.sub.6~C.sub.30 aryl group and C.sub.5~C.sub.30 hetero aryl group, and each of R.sub.1 and R.sub.2 is independently selected from the group consisting of hydrogen, C.sub.1~C.sub.20 alkyl group and C.sub.6~C.sub.30 aryl group.
10. The OLED of claim 1, wherein the electron blocking material is represented by Formula 6:

##STR00162## wherein each of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is polycyclic aryl group.

11. The OLED of claim 10, wherein the electron blocking material is a compound being one of the followings of Formula 7: ##STR00163## ##STR00164## ##STR00165## ##STR00166## ##STR00167## ##STR00168## ##STR00169##

12. The OLED of claim 1, wherein the first hole blocking material is represented by Formula 8: ##STR00170## wherein each of Y.sub.1 to Y.sub.5 are independently CR.sub.1 or N, and one to three of Y.sub.1 to Y.sub.5 is N, wherein R.sub.1 is independently hydrogen or C.sub.6~C.sub.30 aryl group, wherein L is C.sub.6~C.sub.30 arylene group, and R.sub.2 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group, wherein R.sub.3 is hydrogen, or adjacent two of R₃ form a fused ring, and wherein "a" is 0 or 1, "b" is 1 or 2, and "c" is an integer of 0 to 4.

13. The OLED of claim 12, wherein the first hole blocking material is a compound being one of the followings of Formula 9: ##STR00171## ##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176## ##STR00177## ##STR00178## ##STR00179##

14. The OLED of claim 1, wherein the second hole blocking material is represented by Formula 10: ##STR00180## wherein Ar is C.sub.10~C.sub.30 arylene group, R.sub.1 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group, and wherein R.sub.2 is C.sub.1~C.sub.10 alkyl group or C.sub.6~C.sub.30 aryl group.

15. The OLED of claim 14, wherein the second hole blocking material is a compound being one of the followings of Formula 11: ##STR00181##

16. The OLED of claim 1, further comprising: a second emitting material layer including the first host, the second host and the blue dopant and positioned between the first emitting material layer and the second electrode; and a first charge generation layer between the first and second emitting material layers.

17. The OLED of claim 16, further comprising: a third emitting material layer emitting a yellow-green light and positioned between the first charge generation layer and the second emitting material layer; and a second charge generation layer between the second and third emitting material layers.

18. The OLED of claim 16, further comprising: a third emitting material layer emitting a red light and a green light and positioned between the first charge generation layer and the second emitting material layer; and a second charge generation layer between the second and third emitting material layers.

19. An organic light emitting device, comprising: a substrate; and an organic light emitting diode positioned on the substrate and including a first electrode; a second electrode facing the first electrode; a first emitting material layer including a first host, a second host and a blue dopant and positioned between the first and second electrodes; a first electron blocking layer including an electron blocking material of an amine derivative and positioned between the first electrode and the first emitting material layer; and a first hole blocking layer including at least one of a first hole blocking material and a second hole blocking material and positioned between the second electrode and the first emitting material layer, wherein the first host is an anthracene derivative, and the second host is a deuterated anthracene derivative, wherein the first hole blocking material is an azine derivative, and the second hole blocking material is a benzimidazole derivative, wherein the first host is represented by Formula 1: ##STR00182## wherein in Formula 1, R.sub.1 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 heteroaryl group, and R.sub.2 is C.sub.6~C.sub.30 aryl group, and each of L.sub.1 and L.sub.2 is independently C.sub.6~C.sub.30 arylene group, and wherein each of a and b is an integer of 0 or 1, and at least one of a and b is 0, wherein the second host is represented by Formula 3: ##STR00183## wherein in Formula 3, R.sub.1 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 heteroaryl group, and R.sub.2 is dibenzofuranyl or benzo[b]naphthofuranyl, wherein each of L.sub.1 and L.sub.2 is independently C.sub.6~C.sub.30

arylene group, and each of a and b is an integer of 0 or 1, and at least one of a and b is 0, and wherein each of x, y, m and n is independently a positive integer, and a summation of x, y, m and n may be 15 to 29.

20. The organic light emitting device of claim 19, wherein the first host is a compound being one of the followings of Formula 2: ##STR00184## ##STR00185## ##STR00186##

21. The organic light emitting device of claim 19, wherein a weight % ratio of the first host to the second host is 1:9 to 9:1.

22. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 1:9 to 7:3.

23. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 3:7.

24. The organic light emitting device of claim 21, wherein the weight % ratio of the first host to the second host is 7:3.

25. The organic light emitting device of claim 19, wherein the first blue dopant is represented by Formula 4 or Formula 5: ##STR00187## wherein in Formula 4, each of c and d is independently an integer of 0 to 4, and e is an integer of 0 to 3, wherein each of R.sub.11 and R.sub.12 is independently selected from the group consisting of C.sub.1~C.sub.20 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.6~C.sub.30 aryl amino group, or adjacent two among R.sub.11 or adjacent two among R.sub.12 form a fused aromatic ring or a hetero-aromatic ring, wherein R.sub.13 is selected from the group consisting of C.sub.1~C.sub.10 alkyl group, C.sub.6~C.sub.30 aryl group, C.sub.5~C.sub.30 hetero aryl group and C.sub.5~C.sub.30 aromatic amino group, wherein each of X.sub.1 and X.sub.2 is independently oxygen (O) or NR.sub.14, and R.sub.14 is C.sub.6~C.sub.30 aryl group, and wherein in Formula 5, each of Ar.sub.1, Ar.sub.2, Ar.sub.3 and Ar.sub.4 is independently selected from the group consisting of C.sub.6~C.sub.30 aryl group and C.sub.5~C.sub.30 hetero aryl group, and each of R.sub.1 and R.sub.2 is independently selected from the group consisting of hydrogen, C.sub.1~C.sub.20 alkyl group and C.sub.6~C.sub.30 aryl group.

26. The organic light emitting device of claim 19, wherein the electron blocking material is represented by Formula 6: ##STR00188## wherein each of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is independently selected from the group consisting of monocyclic aryl group or polycyclic aryl group, and at least one of R.sub.1, R.sub.2, R.sub.3 and R.sub.4 is polycyclic aryl group.

27. The organic light emitting device of claim 26, wherein the electron blocking material is a compound being one of the followings of Formula 7: ##STR00189## ##STR00190## ##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195##

28. The organic light emitting device of claim 19, wherein the first hole blocking material is represented by Formula 8: ##STR00196## wherein each of Y.sub.1 to Y.sub.5 are independently CR.sub.1 or N, and one to three of Y.sub.1 to Y.sub.5 is N, wherein R.sub.1 is independently hydrogen or C.sub.6~C.sub.30 aryl group, wherein L is C.sub.6~C.sub.30 arylene group, and R.sub.2 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group, wherein R.sub.3 is hydrogen, or adjacent two of R3 form a fused ring, and wherein "a" is 0 or 1, "b" is 1 or 2, and "c" is an integer of 0 to 4.

29. The organic light emitting device of claim 28, wherein the first hole blocking material is a compound being one of the followings of Formula 9: ##STR00197## ##STR00198## ##STR00199## ##STR00200## ##STR00201## ##STR00202## ##STR00203## ##STR00204## ##STR00205##

30. The organic light emitting device of claim 19, wherein the second hole blocking material is represented by Formula 10: ##STR00206## wherein Ar is C.sub.10~C.sub.30 arylene group, R.sub.1 is C.sub.6~C.sub.30 aryl group or C.sub.5~C.sub.30 hetero aryl group, and wherein R.sub.2 is C.sub.1~C.sub.10 alkyl group or C.sub.6~C.sub.30 aryl group.

31. The organic light emitting device of claim 30, wherein the second hole blocking material is a

compound being one of the followings of Formula 11: ##STR00207##

32. The organic light emitting device of claim 19, wherein the organic light emitting diode further includes: a second emitting material layer including a third host, a fourth host and a second blue dopant and positioned between the first emitting material layer and the second electrode; and a first charge generation layer between the first and second emitting material layers, wherein the third host is an anthracene derivative, and the fourth host is a deuterated anthracene derivative.

33. The organic light emitting device of claim 32, wherein the organic light emitting diode further includes: a third emitting material layer emitting a yellow-green light and positioned between the first charge generation layer and the second emitting material layer; and a second charge generation layer between the second and third emitting material layers.

34. The organic light emitting device of claim 33, wherein a red pixel, a green pixel and a blue pixel are defined on the substrate, and the organic light emitting diode corresponds to each of the red, green and blue pixels, and wherein the organic light emitting device further includes: a color filter layer disposed between the substrate and the organic light emitting diode or on the organic light emitting diode and corresponding to the red, green and blue pixels.

35. The organic light emitting device of claim 32, wherein the organic light emitting diode further includes: a third emitting material layer emitting a red light and a green light and positioned between the first charge generation layer and the second emitting material layer; and a second charge generation layer between the second and third emitting material layers.

36. The organic light emitting device of claim 19, wherein a red pixel, a green pixel and a blue pixel are defined on the substrate, and the organic light emitting diode corresponds to each of the red, green and blue pixels, and wherein the organic light emitting device further includes: a color conversion layer disposed between the substrate and the organic light emitting diode or on the organic light emitting diode and corresponding to the red and green pixels.

37. The organic light emitting device of claim 19, wherein the second host is a compound being one of the followings of Formula 12: ##STR00208## ##STR00209## ##STR00210##

##STR00211## ##STR00212## ##STR00213## ##STR00214## ##STR00215## ##STR00216## ##STR00217##
